



Southeast Tech · · Computer Information Systems

Advanced Object Oriented Programmin CIS-249 1C

Fall Term 2020-2021 School Year Section 1C 3.00 Credits 08/24/2020 to 12/11/2020 Modified 08/14/2021

Course Description

Provides students with the knowledge necessary to understand, write and debug two and three tier, object-oriented applications using the Java programming language. Topics covered include Object Oriented Application Development (OOAD), UML (Unified Modeling Language), Web server support for Java, Java Beans, Sockets, Ports, and JDBC (Java Database Connectivity). Also covers accessing DBMS (Database Management Systems) using SQL statements imbedded in Java programs and using HTML web pages combined with JSP's (Java Server Pages). A three tier, database driven business application is developed during the course and deployed for use on both a local area network and for use on the Internet. Prerequisite: CIS 149

Outcomes

Course Outcomes

Upon successful completion of this course, the student shall be able to

- Define and utilize database terminology effectively.
- Create queries using SQL.
- Diagram users' requirements using modeling techniques.
- Transform data models into normalized relations.
- Describe database access standards.

Program Outcomes

Science and Technology

- Develop software applications to solve a business problem.
- Demonstrate proficiency in the design and use of databases.

Problem Solving/Critical Thinking

- Appraise the strengths and weaknesses of one's completed work.
- Analyze a client's software requirements in order to develop possible solutions.

Professionalism

- Utilize professional resources (i.e. professional organizations, media and IT websites).
- Use effective time management skills.

Communication

- Research and present an IT topic using appropriate terminology.
- Write programs with clarity, efficiency, and accuracy.

Additional Outcomes

Course Materials

Oracle online curriculum is used in this course which also serves as the textbook.

Deliverables

Assignments

Assignments are to be turned in via MyTech while quizzes and tests are to be completed online in the Oracle online curriculum.

Evaluation Procedures and Grading

Criteria

Exams (40% of grade)

Exams are assigned from the Oracle web site. There are 6 exams (2 1-part and 2 2-part). The lowest test score will be dropped from the final grade.

Quizzes (20% of grade)

Quizzes are given after most lessons. They are being recorded in 4 parts (midterm, final, midterm, final)

Labs/Database Projects (40% of grade)

- Each assignment will be given a due date.
- Being absent from class is not an excuse for turning in an assignment late.
- A due date will not be changed unless prior arrangements have been made.
- Student may receive partial credit for partial solutions.

Breakdown

Grading Scale:

A+ 99-100%	B+ 89%	C+ 79%	D + 69%	F = 59% and Below
A 94-98%	B 84-88%	C 74-78%	D 64-68%	
A- 90-93%	B- 80-83%	C- 70-73%	D- 60-63%	

Additional Items

The class is broken into two parts

Part 1:

Database Design

This part lays the foundation for understanding relational databases and database design. After learning about the Oracle Academy and Oracle's commitment to education, students plunge into the theory and practice of data modeling. A data model is a conceptual representation of the data structures in a database. The data structures include the data objects (entities), the associations between data objects (relationships), and the rules that dictate operations on the objects (business rules). Students are exposed to data modeling through many real life examples that are relevant to their age group -- case studies include a fast food restaurant and a small music business. In addition to data modeling concepts, the course engages students in activities designed to develop "professional skills." The skills highlighted in the Database Design course include teamwork and team building, public speaking, interviewing, presentation, and project management. The data modeling portion of the course culminates in Section 15 with a final project which assesses all of the data modeling skills learned to date. After completing the project, students begin to learn the Structured Query Language (SQL), the standard language for querying and modifying relational databases. The final three sections of the course (Sections 16-18) introduce the SQL programming language and teach students how to write basic SELECT statements

Daily assignments will be completed in a group setting.

Part 2:

Database Programming with SQL

In this part students gain an understanding of relational databases through the powerful Structured Query Language (SQL). The SQL commands, functions, and operators supported by Oracle as extensions to standard SQL are emphasized. Students learn to create and maintain database objects such as tables, indexes, views, constraints, and sequences. Students practice SQL using SQL Workshop, an application that is available via a web browser in Oracle HTML DB. The Database Programming with SQL portion of the Academy curriculum is designed to help prepare students to pass the Oracle Certified Professional (OCP) exam, "Introduction to Oracle9i SQL Exam." As part of

the Academy, many students will want to take the OCP exam as it is the industry standard for proficiency with SQL. In the Database Programming with SQL course, students focus on searching for a job, job interviewing skills, and writing cover and thank you letters.

Daily Assignments will be completed as individuals.



Course Outline
